

Progress Report: Modelling and Analysis of Multi-Messenger Blazar Spectra

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1 Background and Motivation

1.1 IceCube, Neutrinos, and Blazars

The origin of the diffuse astrophysical neutrino flux observed by the IceCube Neutrino Observatory remains an open question in high-energy astrophysics. After over a decade and a half of operation, few strong candidates for point sources have been identified. The most strongly evidenced are two active galactic nuclei (AGN): the blazar TXS 0506+056 and the Seyfert galaxy NGC 1068. TXS 0506+056 is of particular interest as a multi-messenger source, as its detected neutrino emission coincided with a period of elevated gamma-ray activity observed by the Fermi Large Area Telescope (LAT)(Keivani 2018), providing the first plausible association between a high-energy neutrino event and an astrophysical source. However, stacking analyses have placed firm upper limits on the collective contribution of the blazar population to the diffuse flux at 10–30%, depending on assumptions about the correlation between gamma-ray and neutrino emission. (Aartsen et al. 2017)

Despite these constraints, blazars remain a compelling area of study for multi-messenger research. Their relativistic jets produce some of the most luminous broadband emission in the universe, with particles accelerated to ultra-high energies providing natural conditions for both photon and neutrino production. Understanding the physical parameters that govern jet emission is relevant both to cosmology and to particle physics.

In this project, we approach this problem through the lens of machine learning: Given a large catalogue of simulated blazar spectral energy distributions (SED), for both neutrinos and photons, along with the physical parameters used to generate them, we ask: Can we train models to recover parameters from just the SED? And can this be extended to observationally realistic scenarios, such as heavily masked or incomplete photon SEDs?

1.2 Emission Processes and Jet Mechanisms in Blazars

Blazars are AGN whose relativistic jets are oriented at a small angle θ to the observer's line of sight. Emission produced in the comoving (jet) frame is transformed into the observer frame by the relativistic Doppler factor

$$\delta = \frac{1}{\Gamma(1 - \beta \cos \theta)}, \quad (1)$$

where $\Gamma = (1 - \beta^2)^{-1/2}$ is the bulk Lorentz factor of the jet and $\beta = v/c$ is the corresponding velocity in units of the speed of light. For blazars, $\theta \sim 1/\Gamma$, so that $\delta \sim \Gamma$ and Doppler boosting is extreme, with the boosted luminosity scaling as δ^4 ; a large part of why blazars appear among the most luminous persistent sources in the sky despite comparatively modest intrinsic jet powers. (Urry and Padovani 1995)

Blazars divide broadly into two subclasses, BL Lacertae objects (BL Lacs) and flat-spectrum radio quasars (FSRQs), observationally distinguished by the strength of their external photon fields (accretion disk, broad-line region, and dusty torus), which serve as targets for both inverse Compton scattering by electrons and photopion interactions with protons. This results in different emission line widths and strengths between the two classes (Ghisellini 2013). Regardless of subclass, the broadband SED typically shows two humps: a synchrotron peak from relativistic electrons gyrating in the jet's magnetic field, and a higher-energy peak from inverse Compton scattering (or, in hadronic models, a contribution from proton-induced cascades).

The physical link that motivates this project is that the same relativistic protons responsible for a hadronic gamma-ray contribution also produce high-energy neutrinos via photopion interactions with the ambient photon field, with charged pion decay ultimately yielding neutrinos carrying roughly $E_\nu \approx E_p/20$ (Waxman and Bahcall 1997), while the accompanying neutral pion decay ($\pi^0 \rightarrow 2\gamma$) injects photons into the same cascade. This coupling between the photon and neutrino spectra is the multi-messenger connection this project exploits: a source's observable photon SED carries an imprint of whether, and how strongly, it should also be a neutrino emitter, which is precisely the mapping we aim to learn. However, this relationship is non-trivial, since the dense photon field that acts as a target for photopion interaction will also absorb hadronic gamma-rays, thereby suppressing the correlation and shifting gamma-ray flux to lower energies, depending on the opacity of the source.

1.3 Numerical Modelling with AM^3

We model blazar emission using AM^3 (Astrophysical Multi-messenger Modelling package)(Klinger et al. 2024). This time-dependent, self-consistent, numerical model solves the integro-differential equations describing the coupled evolution of protons, electrons, photons, pions, muons, and neutrinos in a one-zone model (Gao et al. 2017). AM^3 tracks the full cascade of particles from injection to escape and returns steady-state SEDs of neutrinos and photons.

2 Work To Date

2.1 SED Generation

The current way we model the blazars works as follows. We have a spherical blob of material moving along the jet axis at a physically motivated but uniformly sampled relativistic speed. We define a broken power-law spectrum for protons and electrons that are being injected into the blob by Fermi acceleration. We also build a spectrum of external photons out of contributions from the accretion disk, broad line region, and the dusty torus that surround the black hole. We define a uniform magnetic field that permeates the blob. We define several simulation parameters, the most important of which are the maximum and minimum energies of the energy grid, as these determine the size of each energy bin in the grid due to the fixed number of bins. We also set a solver time step to be a fraction of the light-crossing time of the blob. We use the solver to numerically evaluate the equations describing the various particle interactions within the blob. This is done for enough time steps to account for at least one light-crossing time and to allow the emission spectrum to reach a steady-state. At this point, we have a neutrino and photon spectrum in the frame of the jet. Next, we apply redshift and EBL attenuation (Primack et al. 2011) to these jet-frame spectra and shift them into the frame of an observer on Earth. We have primarily sampled our parameters from independent, uniform distributions, with upper and lower limits defined by the results of modelling done in (Rodrigues et al. 2024). The current pipeline generates approximately 1 SED per second running locally. In future, for publication datasets, we aim to run our script on more substantial hardware. Currently, our largest set of data is made up of >70,000 SEDs. Each SED is tagged with the sample parameters used to generate it.

2.2 Parameter Regression

We chose to use Scikit-learn’s gradient-boosted histogram model (Pedregosa et al. 2011) to regress our parameters. The model was selected for its speed on large datasets. To prepare our data for regression, we interpolate our photon spectra onto the same energy grid to simplify the feature vector, and apply mean subtraction normalisation. We have attempted to regress on randomly masked SEDs with both uniform and flux-weighted masking. We have also attempted regression on concatenated neutrino and photon SEDs, though this is highly unrealistic as we do not expect to ever be able to recover the neutrino SED for a source from Earth.

Our results for unmasked regression on photonic SEDs show promise for reconstructing redshift, Γ ; the bulk Lorentz factor of the blob, magnetic field strength, and the total luminosities of injected protons and electrons. We have had weaker results with the maximum energy of injected hadrons, and have been unable to return any results for the maximum injected electron energy. The quality of results begins to fall drastically for SEDs masked by up to 20%. For the concatenated neutrino and photon SEDs, our regression becomes near perfect for all parameters, apart from the leptonic maximum energy, which remains poorly constrained even with the neutrino spectrum available as an additional feature. See figures 1-3 for a visual illustration of these preliminary results.

2.3 Blazar Subclass Classification

Alongside parameter regression, we treat the presence of a significant hadronic component and of a significant external photon field as two independent binary labels, giving four possible source categories: hadronic with external photons, hadronic without, non-hadronic with external photons, and non-hadronic without. These labels are derived from continuous quantities in our simulation (the degree to which hadronic luminosity and external photon flux are suppressed relative to their unsuppressed values) by applying a threshold, rather than being sampled as discrete choices, so that the same underlying dataset can support both the regression task above and this classification task without being generated twice.

We use the same gradient-boosted approach as for regression, adapted to a 4-class classification problem, and evaluate performance with a normalised confusion matrix, which shows the fraction of each true class that is recovered as each predicted class. Early results suggest the model separates the hadronic/non-hadronic axis more reliably than the external-photon-field axis, indicating some degeneracy between the effects of our parameter space and the strength of injected photons. This is an active area of investigation rather than a finished result.

2.4 IceCube Sensitivity

Given that we have the observer-frame neutrino SEDs for our simulated sources, the next obvious step is to investigate which of these source classes IceCube could detect neutrinos from. For a source at declination δ_{src} with observer-frame differential flux $\Phi(E)$, the

Figure 1

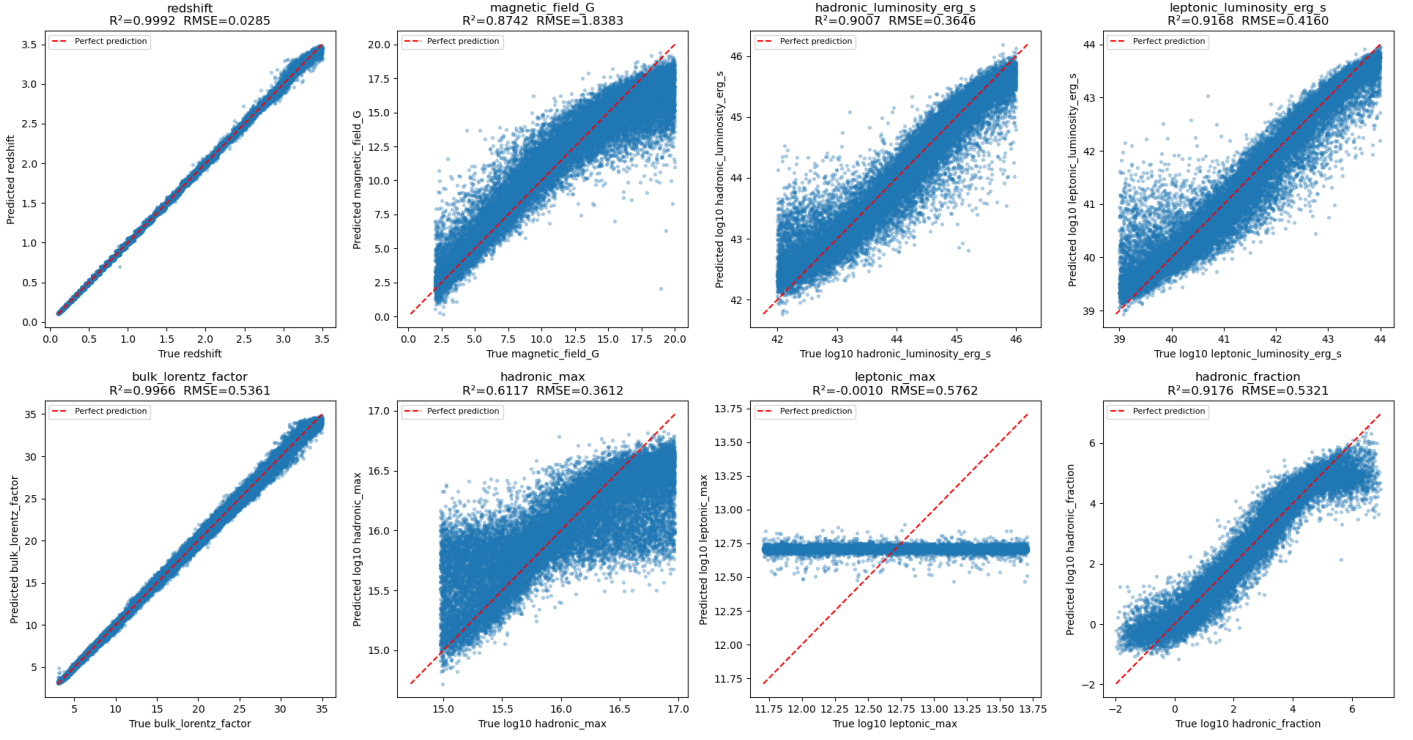


Figure 1: Regression with photon SEDs

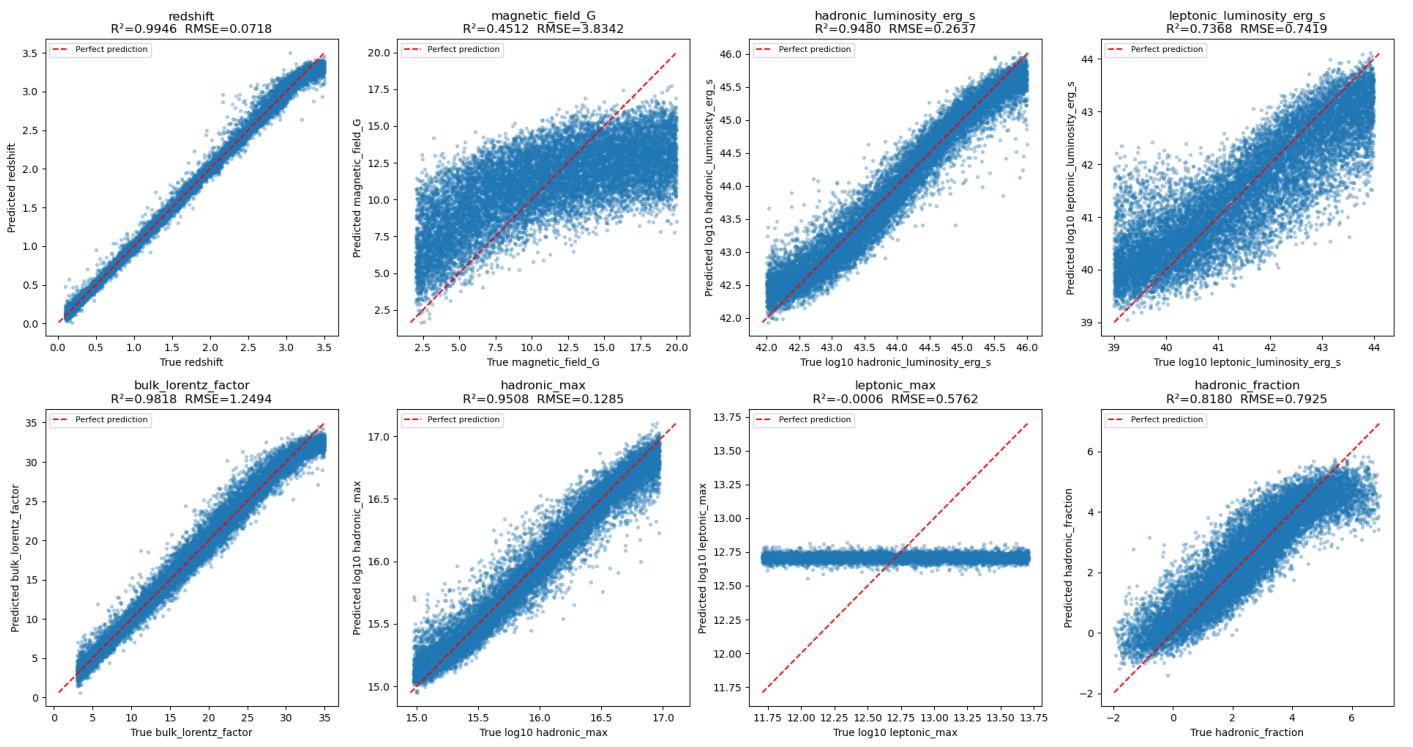


Figure 2: Regression Results with Photon and Neutrino SEDs

Figure 1

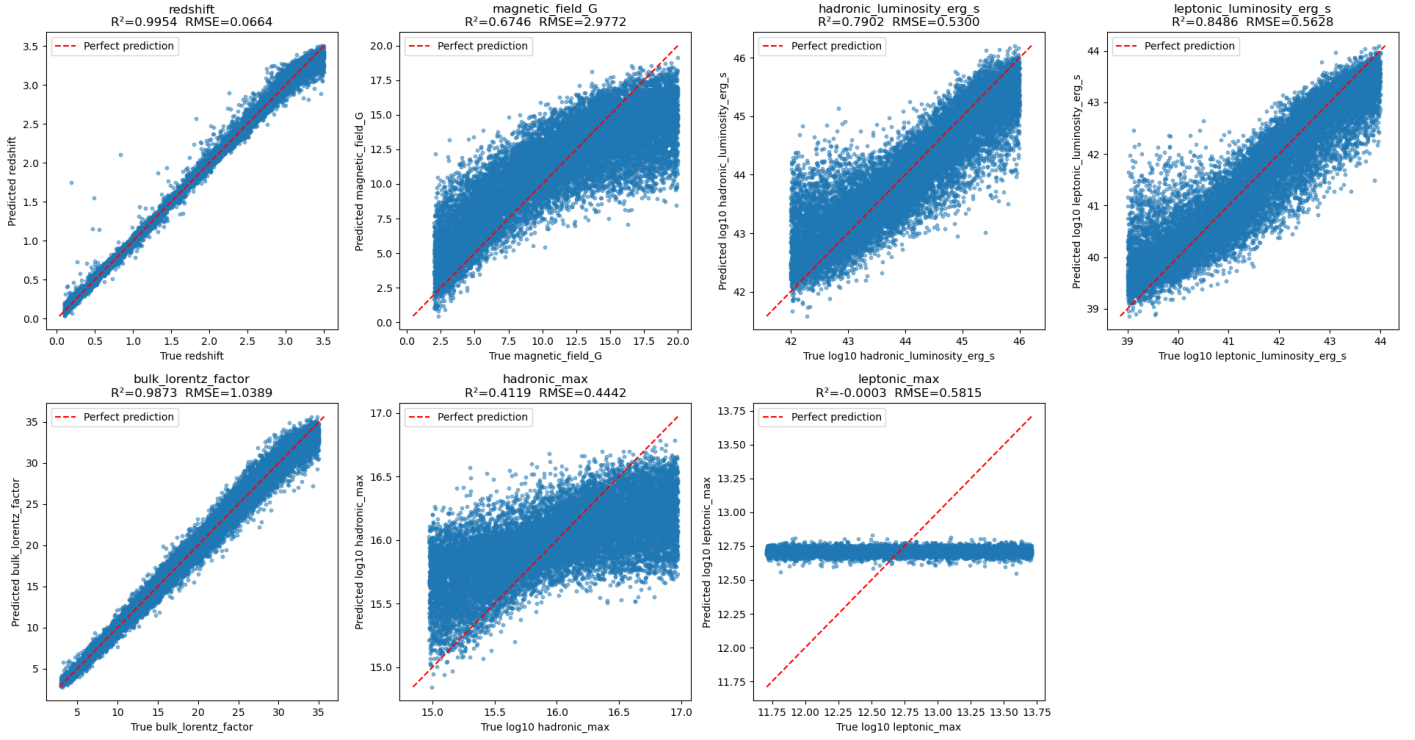


Figure 3: Parameter Regression with 20% masked photon SEDs

expected number of signal events accumulated by IceCube over an exposure time T is

$$N_{\text{sig}}(\delta_{\text{src}}) = T \int dE \Phi(E) A_{\text{eff}}(E, \delta_{\text{src}}), \quad (2)$$

where $A_{\text{eff}}(E, \delta_{\text{src}})$ is the declination- and energy-dependent effective area of the detector. Because A_{eff} varies by orders of magnitude with declination (owing to Earth absorption of upward-going neutrinos and differing background rates), the same intrinsic source flux is far easier to detect at some sky positions than others.

To assess whether a simulated source is detectable, we compute the equivalent expected number of events, $N_{\text{sens}}(\delta_{\text{src}})$, that would be produced by a source emitting at IceCube's published point-source sensitivity flux (Abbasi et al. 2026), $\Phi_{\text{sens}}(E, \delta_{\text{src}})$, evaluated using the same effective area:

$$N_{\text{sens}}(\delta_{\text{src}}) = T \int dE \Phi_{\text{sens}}(E, \delta_{\text{src}}) A_{\text{eff}}(E, \delta_{\text{src}}). \quad (3)$$

The ratio of these two quantities defines a dimensionless detectability statistic,

$$\mathcal{D}(\delta_{\text{src}}) = \frac{N_{\text{sig}}(\delta_{\text{src}})}{N_{\text{sens}}(\delta_{\text{src}})}, \quad (4)$$

with $\mathcal{D} \geq 1$ indicating that, were the source located at declination δ_{src} , its flux would meet or exceed the threshold IceCube requires for a point-source detection. Since a source's true declination is fixed but we wish to characterise its detectability in a way that is not tied to one particular, possibly unfavourable, sky position, we average $\mathcal{D}$ over declination,

$$\bar{\mathcal{D}} = \langle \mathcal{D}(\delta_{\text{src}}) \rangle_{\delta_{\text{src}}} = \int d\delta_{\text{src}} w(\delta_{\text{src}}) \mathcal{D}(\delta_{\text{src}}), \quad (5)$$

where $w(\delta_{\text{src}})$ is a weighting that can be taken as uniform in $\sin \delta_{\text{src}}$ or restricted to the visible sky as appropriate. In effect, this procedure identifies sources whose bolometric neutrino flux exceeds the IceCube detection threshold for at least some fraction of sky positions, allowing us to rank our simulated source population by observational detectability independently of where any individual source happens to sit on the sky.

The results of this filtering are telling when we examine the distribution of the parameters of the surviving sources. We see that low redshift and high Γ , hadronic luminosity, and maximum hadronic energy are determinants of detection via neutrinos. These results align with the assumption that the most detectable neutrino sources are those that are nearest to us with the highest Doppler boosting and hadronic contribution. See Figure 4

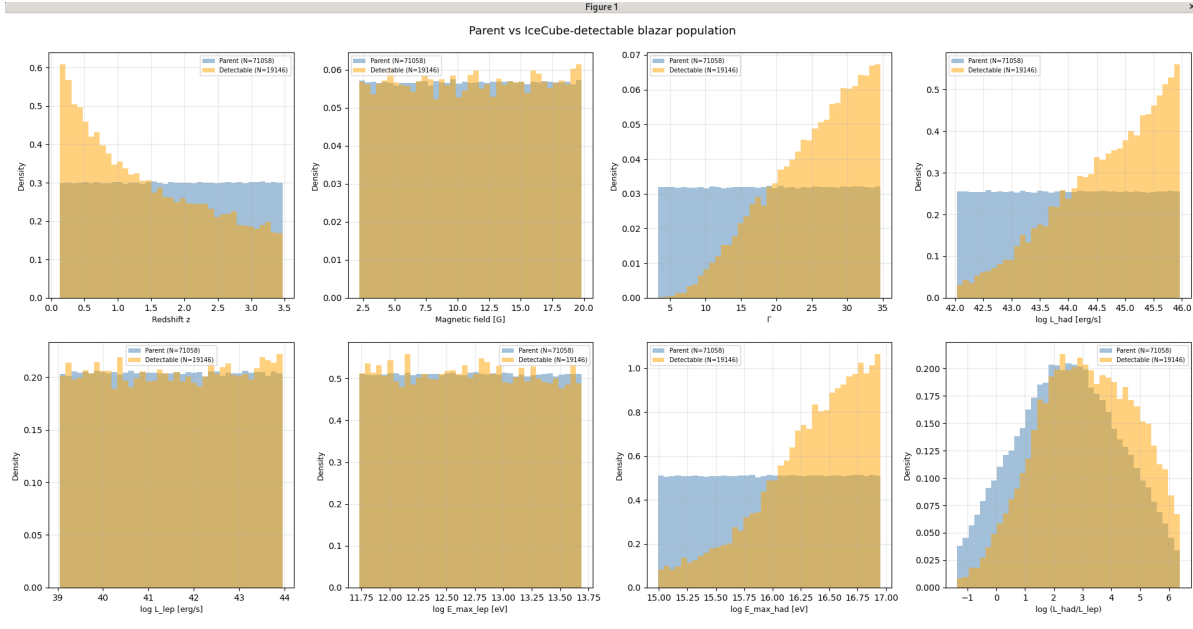


Figure 4: Parameter distributions of the parent blazar population compared to the subset identified as detectable by IceCube under the filtering procedure of Equation 5.

3 Difficulties and Future Work

Our current results are very preliminary. We are currently interested in developing a robust generation pipeline and developing our machine learning techniques, with more work still to be done on parameter distributions, dimensionality reduction, detectability modelling, and the points below.

One of the key points of struggle throughout the work so far is deciding on an effective way to sample parameters to build our dataset of blazar SEDs. We have an incomplete understanding of the physical processes that lead to the properties of the injection spectra. As such, we have struggled with how to best preserve any underlying correlations between parameters. Being able to resolve this issue would be of great benefit, as it would restrict the data we train our models on to only physically relevant cases.

We have since extended our sampling to include a continuous range of hadronic and external photon field strengths, rather than treating their presence as a simple on/off choice, so that our dataset now spans smoothly from strongly suppressed to unsuppressed sources along both axes. A separate, ongoing issue is that a subset of our generated SEDs display only a single spectral peak rather than the expected two-hump structure; we are still isolating whether this reflects a genuine physical regime (for example, a fixed lower cutoff on injected electron energy that may exclude a soft synchrotron component for some parameter combinations) or an artefact of insufficient evolution time in the solver, and resolving this is a priority before further production runs.

Finally, our IceCube detectability statistic carries a caveat worth flagging: published point-source sensitivity fluxes assume a fixed power-law spectral index, whereas our simulated spectra can deviate substantially from that assumed shape. Properly accounting for this would require interpolating between sensitivity curves computed at different spectral indices, rather than comparing our spectra against a single assumed power law.

4 References

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